

MID-SEMESTER EXAMINATION

Course Name: Rational homotopy theory

Time: 2 hr, Date: 26.02.2026

Total Marks: 20

Read the following instructions:

- Answer any four questions.
- All questions carry equal marks unless stated otherwise.
- Give complete proofs and justify all steps. If you are using a result (Theorem/Proposition/Lemma/Corollary) discussed in class, state it clearly.

✓ 1. Prove that $\mathbb{CP}^\infty$ is an Eilenberg–MacLane space of type $K(\mathbb{Z}, 2)$. [5]

2. Define the Betti numbers b_i of a space X by

$$b_i = \dim_{\mathbb{Q}} H^i(X; \mathbb{Q}).$$

Prove that a path connected space X with finite fundamental group has $b_1 = 0$.

- Prove or disprove whether the converse is true?

[3+2]

3. Let M be a closed orientable smooth simply connected three-manifold.

✓(a) Prove that M is homotopy equivalent to S^3 .

(b) Show that for every $n \geq 4$, a simply connected closed n -manifold need not be homotopy equivalent to S^n .

[2.5+2.5]

✓ 4. Prove that the rationalization of $\Sigma\mathbb{CP}^2$ is homotopy equivalent to

$$S_{\mathbb{Q}}^3 \vee S_{\mathbb{Q}}^5.$$

[5]

5. Compute the cohomology ring $H^*(\Omega S^3; \mathbb{Z})$ of the based loop space of the three-sphere. Show that there exists a map

$$\Omega S^3 \longrightarrow \mathbb{CP}^\infty$$

which induces an isomorphism on rational cohomology

$$H^*(-; \mathbb{Q}).$$

[5]

- $\mathcal{K}$. Let $(A, d = 0)$ and $(B, d = 0)$ be two cohomologically 1-connected DGAs with zero differentials and $f : B \rightarrow A$ be a morphism of DGAs. Let $\rho_B : M_B \rightarrow B$ and $\rho_A : M_A \rightarrow A$ be respective minimal models. Show that there exists a map $\tilde{f} : M_B \rightarrow M_A$ such that $\rho_A \circ \tilde{f} \simeq f \circ \rho_B$ (here " $\simeq$ " means homotopic to). Prove or disprove $\rho_A \circ \tilde{f} = f \circ \rho_B$. [1.5+3.5]
- $\mathcal{J}$. Write down the rational cohomology algebra $A = H^*(S^2 \vee S^2; \mathbb{Q})$. Consider $(A, d = 0)$ as a DGA and compute its minimal model up to degree 5. [1+4]

raktim13.github.io